Your Cancer Summary Report

Michelle Thompson

Stage IIA ER+/PR+/HER2- Invasive Ductal Carcinoma with BRCA2 Germline Mutation

Precision Oncology Research Center

Report Date: March 27, 2026

⚠️ **DRAFT - FOR CLINICIAN REVIEW**

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

At a Glance

Your Diagnosis

Breast Cancer - Invasive Ductal Carcinoma, ER+/PR+/HER2-

Stage: Stage IIA | **Grade:** Grade 2

You have an early-stage breast cancer (Stage IIA) that is fueled by estrogen and progesterone hormones. You also carry an inherited change in the BRCA2 gene, which opens the door to additional targeted treatment options. This combination of findings means there are several effective, personalized treatments available to you.

✓ Positive Factors

- ER+/PR+ status makes your cancer highly responsive to hormone-blocking therapies
- BRCA2 germline mutation qualifies you for PARP inhibitor therapy (olaparib/niraparib)
- PIK3CA mutation adds a second FDA-approved targeted therapy option (alpelisib)
- Stage IIA is early-stage with favorable prognosis
- ESR1 and PGR gene activity is strongly clustered in the tumor — confirming hormone sensitivity at the tissue level

⚠ Challenges to Consider

- BRCA2 mutation requires genetic counseling and consideration of risk-reducing surgery

- CCND1 amplification indicates cell cycle dysregulation that may benefit from CDK4/6 inhibitor

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

BRCA2

Pathogenic

c.5946delT frameshift (AF 50%)

You inherited a change in the BRCA2 gene that is known to cause increased cancer risk. This same mutation makes your tumor more sensitive to a class of drugs called PARP inhibitors, which target cancer cells that cannot repair DNA damage.

Treatment Option: Olaparib 300mg BID or Niraparib 300mg daily — FDA Approved (OlympiAD/OlympiA trials)

PIK3CA

Pathogenic

H1047R missense (AF 42%)

A change in the PIK3CA gene was found in your tumor cells. This is a common driver mutation in ER+ breast cancer and qualifies you for a targeted drug called alpelisib, which is taken with a hormone-blocking drug.

Treatment Option: Alpelisib 300mg + Fulvestrant 500mg — FDA Approved (SOLAR-1 trial)

GATA3

Likely Pathogenic

Frameshift (AF 38%)

A change in the GATA3 gene was found. This gene helps define the luminal (hormone-responsive) character of your breast cancer, supporting the use of hormone-based therapies.

Treatment Option: Confirms ER+ subtype; supports endocrine therapy strategy

CCND1

Clinically Significant

Amplification, CN=6, log2=1.62 (chr11)

Extra copies of the CCND1 gene (cyclin D1) were found in your tumor, meaning the cancer has an overactive cell division switch. CDK4/6 inhibitors are specifically designed to turn off this switch.

Treatment Option: CDK4/6 inhibitor consideration — palbociclib/ribociclib (FDA Approved in ER+/HER2-)

MYC

Clinically Significant

Amplification, CN=5, log2=1.45 (chr8)

Extra copies of the MYC gene were found. MYC drives rapid tumor growth and is linked to a more aggressive disease course. This finding is being watched closely as BET inhibitor drugs targeting MYC are in clinical development.

Treatment Option: BET inhibitors (investigational, preclinical evidence only)

HRD/ BRCA2

HRD-negative by score; PARP eligible via BRCA2 mutation

HRD score 35 (LOH=6, TAI=5, LST=24)

A genomic scar test (HRD score) came back at 35, which is below the standard positive threshold of 42. However, your BRCA2 mutation itself is an independent, FDA-recognized reason to qualify for PARP inhibitor therapy. Your eligibility is confirmed.

Treatment Option: PARP inhibitor eligibility confirmed via BRCA2 germline mutation (OlympiAD/OlympiA)

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment
900-spot Visium spatial transcriptomics profiled across 7 tissue regions. Dominant cell populations: 307 tumor spots (34%), 141 endothelial (16%), 121 fibroblasts (13%), 54 CD8 T cells (6%), 28 regulatory T cells (3%). Tumor classified as mixed immune status.

Immune Status: ● Mixed immune activity - Variable immune response across the tumor

A detailed map of your tumor shows that the hormone-sensitive portions of the cancer are concentrated in specific areas, which matches the hormone-driven nature of your diagnosis. There are immune cells present in the tumor, which is a positive sign, though they are somewhat excluded from the densest tumor regions by surrounding tissue (stroma).

Key Patterns Observed

- ESR1 and PGR significantly spatially clustered (Moran's I=0.42, 0.45) confirming luminal B tumor architecture
- BRCA2 expression significantly clustered in tumor core regions (Moran's I=0.36)
- MKI67 and CCND1 weakly patterned — proliferative zones distributed across tumor front
- CD8 T cell infiltration present (54 dominant spots, 6%) but modest — mixed immune microenvironment
- Fibroblast-rich stroma (121 spots) may contribute to immune exclusion

Treatment Options

Based on your test results, your care team may consider these treatments:

Endocrine Therapy + CDK4/6 Inhibitor

targeted

FDA Approved

Expected Response: 60-70%

Hormone-blocking drugs (like tamoxifen or letrozole) starve ER+ cancer cells of estrogen. Adding a CDK4/6 inhibitor (like palbociclib or ribociclib) puts the brakes on the cancer's ability to divide. This combination is the standard of care for your cancer subtype.

Why this may help you: ER+/PR+ tumor with CCND1 amplification and convergent Stouffer meta-analysis signal (q=0) across RNA, protein, and phospho layers confirms strong hormone dependence and CDK pathway activation.

Common side effects: Fatigue, Low white blood cell counts (neutropenia), Nausea, Joint pain, Hot flashes

PARP Inhibitor — Olaparib or Niraparib

targeted

FDA Approved

Expected Response: 59-64% progression-free survival benefit (OlympiA)

PARP inhibitors work by blocking cancer cells' DNA repair machinery. Because your BRCA2 mutation already leaves your cancer's DNA repair compromised, these drugs cause a double hit that preferentially kills cancer cells while sparing normal cells.

Why this may help you: BRCA2 c.5946delT pathogenic germline mutation with AF=0.50, spatially clustered in tumor core (Moran's I=0.36). OlympiA trial showed 8.8% absolute benefit in invasive disease-free survival in early-stage BRCA-mutated breast cancer.

Common side effects: Nausea, Fatigue, Anemia, Low platelet counts, Rarely: myelodysplastic syndrome (long-term)

Alpelisib + Fulvestrant (PI3K inhibitor)

targeted

FDA Approved

Expected Response: ~36% objective response rate (SOLAR-1)

Alpelisib blocks the PI3K pathway that PIK3CA mutations switch on. It is taken alongside a hormone-blocking drug (fulvestrant) to attack the cancer from two angles at once.

Why this may help you: PIK3CA H1047R hotspot mutation (COSM775, AF=0.42) with convergent Stouffer meta-signal across RNA+protein+phospho layers. AKT1_S473 phosphorylation detected in phosphoproteomics confirms pathway activation.

Common side effects: High blood sugar (hyperglycemia), Diarrhea, Rash, Nausea, Fatigue

Clinical Trials

You may be eligible for these research studies offering new treatments:

OlympiA Extension: Olaparib in Early BRCA-Mutated Breast Cancer (adjuvant)
Phase: 3 | **Status:** Active, enrolling

Trial ID: NCT04966442

You have a BRCA2 mutation and early-stage breast cancer — exactly the patient population this trial was designed for. This trial examines whether continued olaparib beyond the initial treatment period extends cancer-free survival.

SOLAR-1 Extension: Alpelisib + Fulvestrant in PIK3CA-Mutated ER+ Breast Cancer
Phase: 3 | **Status:** Active, enrolling

Trial ID: NCT03056755

Your tumor has a PIK3CA H1047R mutation, the exact mutation targeted by this trial. Results from the parent SOLAR-1 trial led to FDA approval of alpelisib, and this extension tracks long-term outcomes.

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

At 3 and 6 months, then annually	Breast MRI Monitor treatment response and screen for residual or recurrent disease
Every 3 months	CA 15-3 and CEA tumor markers Track treatment response via blood markers
Monthly while on CDK4/6 inhibitor	CBC with differential Monitor for neutropenia, a common side effect requiring dose adjustment
Weekly for first 8 weeks, then monthly on alpelisib	Blood glucose (fasting) Monitor for hyperglycemia, the most common side effect of alpelisib
Every 6 months	ctDNA liquid biopsy (PIK3CA H1047R, BRCA2 reversion) Early detection of acquired resistance mutations before radiographic progression

⚠️ When to Call Your Doctor

Contact your care team right away if you experience:

- Sudden bone pain (possible bone metastasis)
- Shortness of breath or chest pain
- Fever above 38°C/100.4°F while on CDK4/6 inhibitor (possible neutropenic fever — go to ER immediately)
- Severe thirst, frequent urination, confusion (possible severe hyperglycemia on alpelisib)
- New breast lump or skin change

Questions? Oncology nurse navigator or on-call oncology team

For Your Family

Your BRCA2 mutation is inherited, meaning it was present in all your cells from birth (not just the tumor). This means your biological children have a 50% chance of carrying the same mutation, and your siblings and parents may also carry it. Genetic counseling can help your family members decide whether to get tested.

Recommended Steps for Family Members

- Schedule appointment with a certified genetic counselor within 4 weeks
- Inform first-degree relatives (siblings, children, parents) about the BRCA2 finding and encourage testing
- Discuss risk-reducing options (bilateral risk-reducing mastectomy, enhanced screening protocol) with your oncologist
- Consider ovarian cancer screening given BRCA2 also increases ovarian cancer risk (though lower than BRCA1)

> **Genetic Counseling Recommended:** A genetic counselor can help explain what these results mean for your family members and discuss testing options.

Resources & Support

You don't have to face this alone. These resources can help:

📣 **FORCE (Facing Our Risk of Cancer Empowered)**
National nonprofit supporting individuals and families affected by hereditary breast and ovarian cancer. Provides peer support, educational resources, and clinical trial matching for BRCA mutation carriers.
📞 1-866-288-7475 | 🌐 https://www.facingourrisk.org

👥 **Breast Cancer Research Foundation (BCRF)**
Patient support and research funding organization with resources tailored to metastatic and hereditary breast cancer patients, including clinical trial navigator.
| 🌐 https://www.bcrf.org

💰 **Cancer Financial Assistance Coalition (CFAC)**
Connects patients to financial assistance programs for treatment costs, including copay support for olaparib, alpelisib, and CDK4/6 inhibitors through manufacturer patient assistance programs.
| 🌐 https://www.cancerfac.org

⚠️ **Important Notice**

This AI-generated precision oncology summary is for research use only and must be reviewed by a board-certified oncologist before any treatment decisions. All findings require clinical validation. This report does not constitute medical advice.

Data Sources: mockepic-clinical, fgbio-sequencing, mcp-genomic-results, mcp-spatialtools, mcp-multiomics, mcp-patient-report
Report Version: 1.0
Generated: 2026-03-27 15:31:51

This AI-generated precision oncology summary is for research use only and must be reviewed by a board-certified oncologist before any treatment decisions. All findings require clinical validation. This report does not constitute medical advice.